



CE432: GEOGRAPHICAL INFORMATION SYSTEM
TERM PAPER PRESENTATION

2025-26/II

Demonstration of Geographic Information System (GIS) Functions Using Microsoft Excel

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GROUP 7

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Introduction

- Geographic Information Systems (GIS) integrate hardware, software, and spatial data to capture, analyze, and visualize geographic information. Since their early development by Roger Tomlinson, GIS has evolved into a vital tool in civil engineering, supporting applications such as site planning, transportation optimization, flood mapping, and infrastructure management.
- However, professional platforms like ArcGIS can be complex, expensive, and require advanced hardware, limiting accessibility for students and practitioners. This study explores how Microsoft Excel can serve as a simpler alternative to implement key GIS functions.
- Using real-world coordinate data of Indian states and cities, health data of Ghana, the study demonstrates eleven GIS operations, highlighting Excel's effectiveness as a practical and educational tool for understanding spatial analysis.



ArcGIS



Objective

- Identify and demonstrate eleven fundamental GIS functions using Microsoft Excel.
- Convert mathematical formulations of these functions into Excel-based implementations.
- Develop a working Excel model showcasing all eleven GIS operations with real geographic data.
- Evaluate the strengths and limitations of Excel as a GIS learning and analysis platform.



Dataset Description

Dataset 1: District wise centroids

Source : [Kaggle.com](https://www.kaggle.com) Format: Comma Separated Value (CSV)

State	District	Latitude	Longitude
Andaman and Nicobar	Andaman Islands	12.382571069235	92.822910506415

Dataset 2: Health facilities in Ghana

Source : [Kaggle.com](https://www.kaggle.com) Format: Comma Separated Value (CSV)

Region	District	FacilityName	Type	Town	Ownership	Latitude	Longitude
Ashanti	Adansi South	Hwidiem Community Clinic	Clinic	Hwidiem	Government	5.89432	-1.46766

Dataset Description

Dataset 3: Indian states,district wise centroid & altitude

Source : [Github.com](https://github.com) Format: Comma Separated Value (CSV)

district	std-code	state	gst-state-code	iso_3166-2	population	rank	latitude	longitude	altitude
South Andaman	3192	Andaman and Nicobar Islands	AN	IN-AN	108058	423	11.62338	92.72648	16.12904

Dataset 4: Indian States wise total traffic Accidents

Source : data.gov.in Format: Comma Separated Value (CSV)

Sl. No.	State	Road Accident s - Cases	Road Accident s - Injured	Road Accident s - Died	Railway Accident s - Cases	Railway Accident s - Injured	Railway Accident s - Died	Railway Crossing Accident s - Cases	Railway Crossing Accident s - Injured	Railway Crossing Accident s - Died	Total Traffic Accidents - Cases	Total Traffic Accidents - Injured	Total Traffic Accidents - Died
1	Andhra Pradesh	21070	21340	8293	1029	0	1037	0	0	0	22099	21340	9330

Function 1

Distance Calculation (Haversine Formula)

Let Point 1 have coordinates (ϕ_1, λ_1) and Point 2 have coordinates (ϕ_2, λ_2) in radians.

$$d = 2R \arcsin \left(\sqrt{\sin^2 \left(\frac{\phi_2 - \phi_1}{2} \right) + \cos(\phi_1) \cos(\phi_2) \sin^2 \left(\frac{\lambda_2 - \lambda_1}{2} \right)} \right)$$

$R = 6371$ km , ϕ represents latitude in radians and λ represents longitude in radians.

Excel Formulas

```
=IFERROR(XLOOKUP(city1, 'district wise centroids'!District_Names, 'district wise centroids'!Lat_coords), "Not Found")  
=2*6371*ASIN(SQRT(SIN((RADIANS(Lat2-Lat1))/2)^2 +  
COS(RADIANS(Lat1))*COS(RADIANS(Lat2))*SIN((RADIANS(Lon2- Lon1))/2)^2))
```

Point 1 City Name	Varanasi	Point 2 City Name	Bhilwara
Latitude (°N) of point 1	25.3605708888888	Latitude (°N) of Point 2	25.4511976775956
Longitude (°E) of Point 1	82.8252966666667	Longitude (°E) of Point 2	74.6394759562841
Distance (in Km)		822.1280153	



Function 2

Select Within Distance

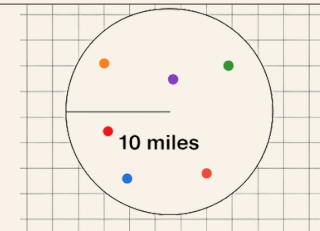
Excel Formulas

```
=IFERROR(XLOOKUP(target_district,District_col,Long_col), "Not Found")
```

```
=COUNTIFS(Distance_from_target, "<=" & proximity_distance, Distance_from_target, ">0")
```

```
=AND(Distance_from_target <= proximity_distance, Distance_from_target>0)
```

```
=FILTER(District_Names, (Distance_from_target <= proximity_distance) *  
(Distance_from_target>0), "None within range")
```



Selection By Proximity	
Name of Target District	Rampur
Latitude of Target District	28.8324472727272
Longitude of Target District	79.1097733333333
Proximity Distance	1229
Total Selected Districts	418

Function 2

Select Within Distance

Results

State	District	Latitude	Longitude	Distance from Target
Andaman and Nicob	Andaman Islands	12.382571069235052	92.82291050641506	2315.51673
Andaman and Nicob	Nicobar Islands	7.835290749774195	93.51160123757859	2779.597686
Andhra Pradesh	Adilabad	19.284513950073435	78.81321233480176	1062.10709
Andhra Pradesh	Anantapur	14.312065561843147	77.46015845	1623.50791
Andhra Pradesh	Chittoor	13.331092630676961	78.92763898886037	1723.774658
Andhra Pradesh	Cuddapah	14.490289134125629	78.69613870967754	1595.343581
Andhra Pradesh	East Godavari	16.782718320784902	82.24320747	1377.613024
Andhra Pradesh	Guntur	15.884925767634849	80.58657614107885	1447.635196
Andhra Pradesh	Hyderabad	17.389715555555556	78.46668888888888	1274.063634
Andhra Pradesh	Karimnagar	18.515483193277312	79.43765428571413	1147.677825
Andhra Pradesh	Khammam	17.683921325051784	80.67008643892335	1249.815548
Andhra Pradesh	Krishna	16.143872753151673	81.14805061000413	1426.265711
Andhra Pradesh	Kurnool	15.43759715142429	77.90086836581717	1494.599653
Andhra Pradesh	Mahbubnagar	16.639546224256293	77.98933089244859	1360.621301
Andhra Pradesh	Medak	17.889304978038055	78.13778901903379	1220.841253
Andhra Pradesh	Nalgonda	17.113250515463907	79.16984453608244	1303.129657
Andhra Pradesh	Nellore	13.894344071421889	80.14702671812498	1664.481882
Andhra Pradesh	Nizamabad	18.429485849056604	78.07523490566035	1161.526764
Andhra Pradesh	Prakasam	15.489039001601705	80.11665790176184	1487.313547
Andhra Pradesh	Rangareddi	17.245235872576178	78.05756828254849	1292.906282
Andhra Pradesh	Srikakulam	18.570900998697432	84.25874814010675	1255.227976
Andhra Pradesh	Vishakhapatnam	17.64683903	83.01014035778171	1305.761508
Andhra Pradesh	Vizianagaram	18.283296960329718	83.45982854198873	1253.660196
Andhra Pradesh	Warangal	18.01752608695651	79.87422028985509	1205.079364
Andhra Pradesh	West Godavari	16.63297943925236	81.45912990654196	1377.628581
Arunachal Pradesh	Changlang	27.360484793388455	96.37552644628106	1699.999416

selected districts
Adilabad
Karimnagar
Medak
Nizamabad
Warangal
Barpeta
Bongaigaon
Dhuburi
Goalpara
Kokrajhar
Araria
Aurangabad
Banka
Begusarai
Bhabua
Bhagalpur
Bhojpur
Buxar
Darbhanga
Gaya
Gopalganj
Jamui
Jehanabad

Function 3

Select by Location

Selection by location is a spatial query operation in GIS that selects features from a target layer that lie within a reference layer

Excel Formulas

```
=AND(Latitudes>= min_latitude, Latitude <= max_latitude, Longitude >= min_longitude, Longitude <= max_longitude)
```

```
=FILTER(Districts, (Latitude>=min_latitude) * (Latitude <=max_latitude) * (Longitude>=min_longitude) * (Longitude<=max_longitude), "None in area")
```

Select by location	
Min Latitude	13
Max Latitude	20
Min Longitude	70
Max Longitude	95
Total districts lies in this box	76

By using conditional formating feature of excel, our result got highlighted just like in QGIS

Function 3

Select by Location

Results

State	District	Latitude	Longitude	Name of districts whose centriod lies in the box
Andaman and Nicobar	Andaman Islands	12.38257107	92.82291051	Adilabad
Andaman and Nicobar	Nicobar Islands	7.83529075	93.51160124	Anantapur
Andhra Pradesh	Adilabad	19.28451395	78.81321233	Chittoor
Andhra Pradesh	Anantapur	14.31206556	77.46	Cuddapah
Andhra Pradesh	Chittoor	13.33109263	78.92763899	East Godavari
Andhra Pradesh	Cuddapah	14.49028913	78.69613871	Guntur
Andhra Pradesh	East Godavari	16.78271832	82.24	Hyderabad
Andhra Pradesh	Guntur	15.88492577	80.58657614	Karimnagar
Andhra Pradesh	Hyderabad	17.38971556	78.46668889	Khammam
Andhra Pradesh	Karimnagar	18.51548319	79.43765429	Krishna
Andhra Pradesh	Khammam	17.68392133	80.67008644	Kurnool
Andhra Pradesh	Krishna	16.14387275	81.14805061	Mahbubnagar
Andhra Pradesh	Kurnool	15.43759715	77.90086837	Medak
Andhra Pradesh	Mahbubnagar	16.63954622	77.98933089	Nalgonda
Andhra Pradesh	Medak	17.88930498	78.13778902	Nellore
Andhra Pradesh	Nalgonda	17.11325052	79.16984454	Nizamabad
Andhra Pradesh	Nellore	13.89434407	80.14702672	Prakasam
Andhra Pradesh	Nizamabad	18.42948585	78.07523491	Rangareddi
Andhra Pradesh	Prakasam	15.489039	80.1166579	Srikakulam
Andhra Pradesh	Rangareddi	17.24523587	78.05756828	Vishakhapatnam
Andhra Pradesh	Srikakulam	18.570901	84.25874814	Vizianagaram
Andhra Pradesh	Vishakhapatnam	17.65	83.01014036	Warangal
Andhra Pradesh	Vizianagaram	18.28329696	83.45982854	West Godavari

Function 4

Nearest Neighbor Analysis

$$d_i = d_{\text{Haversine}}(\text{Query}, P_i)$$

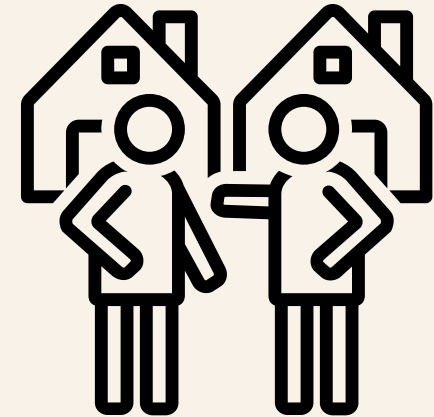
$$P_{\text{nearest}} = \arg \min_i d_{\text{Haversine}}(\text{Query}, P_i)$$

Excel Formulas

=IFERROR(XLOOKUP(target_district,District_col,Long_col), "Not Found")

Result

Nearest Neighbour	
City	Bhilwara
Latitude	25.451198
Longitude	74.639476
Nearest Neighbour	Rajsamand



Function 5

Slope Calculation from Elevation Data

$$\text{Slope (\%)} = \left(\frac{\Delta \text{Elevation}}{\text{Horizontal Distance}} \right) \times 100$$

$$\text{Slope (degrees)} = \tan^{-1} \left(\frac{\Delta \text{Elevation}}{\text{Horizontal Distance}} \right)$$

Excel Formulas

```
= (Z2-Z1)/(Dist_km*1000)*100  
=DEGREES(ATAN((Z2-Z1)/(Dist_km*1000)))
```

Slope w.r.t City	
City	Rampur
Latitude	28.789304
Longitude	79.024952
Altitude	190.06732

Function 5

Slope Calculation from Elevation Data

Result

city	latitude	longitude	altitude	Distance	Slope(%)	Slope(degree)	
Port Blair	11.62338	92.72648	16.12904	2380.097502	0.007308	3.76752E-05	
Adoni	15.63222	77.27284	425.7995	1474.004459	-0.0159926	-6.08933E-05	
Amaravati	16.57452	80.35901	34.70644	1365.061311	0.0113812	6.56609E-05	
Anantapur	14.68189	77.60059	348.9019	1575.505635	-0.0100815	-5.68956E-05	
Bhimavaram	16.54471	81.52059	8.032158	1385.269376	0.0131408	6.47421E-05	
Chilakaluripet	16.09243	80.16239	34.29998	1416.629494	0.0109956	6.32714E-05	
Chittoor	13.21718	79.10033	310.4805	1731.559164	-0.006954	-5.17015E-05	
Dharmavaram	14.42273	77.72147	366.4323	1603.119671	-0.0110014	-5.59379E-05	
Eluru	16.71066	81.09524	21.15529	1359.664612	0.0124231	6.59433E-05	
Gudivada	16.44103	80.99263	12.98032	1387.754245	0.0127607	6.46198E-05	
Guntakal	15.16741	77.37362	456.1523	1524.159437	-0.0174578	-5.89077E-05	
Guntur	16.30665	80.43654	39.71604	1395.511633	0.0107739	6.42194E-05	
Hindupur	13.82907	77.49184	628.6644	1671.003899	-0.0262475	-5.37817E-05	
Kadapa	14.46735	78.82413	137.4309	1592.662418	0.0033049	5.58258E-05	
Kakinada	16.98906	82.24746	7.053235	1352.794877	0.0135286	6.62975E-05	
Kurnool	15.82813	78.03728	284.9749	1444.771333	-0.006569	-6.18758E-05	
Machilipatnam	16.19055	81.13615	10.91093	1417.508035	0.0126388	6.32661E-05	
Madanapalle	13.56035	78.50361	690.9739	1694.237745	-0.0295653	-5.30537E-05	
Nandyal	15.47857	78.48309	211.9017	1481.130986	-0.0014742	-5.89939E-05	
Narasaraopet	16.23535	80.0479	76.20676	1399.861309	0.0081337	6.39326E-05	
Nellore	14.4426	79.98646	19.11972	1598.350503	0.0106953	5.60984E-05	
Ongole	15.50572	80.04992	19.72562	1480.808838	0.0115033	6.05505E-05	
Proddatur	14.74919	78.55316	159.6724	1561.944234	0.001946	5.64141E-05	
Rajamahendravaram	17.00054	81.80403	36.38979	1341.24129	0.0114579	6.6824E-05	

Function 6

Bearing / Azimuth Calculation

$$\theta = \tan^{-1} \left(\frac{\sin(\Delta\lambda) \cos \phi_2}{\cos \phi_1 \sin \phi_2 - \sin \phi_1 \cos \phi_2 \cos(\Delta\lambda)} \right)$$

$$\text{Bearing} = \left(\frac{180}{\pi} \theta + 360 \right) \bmod 360$$



Formulas Used

```
=MOD(DEGREES(ATAN2(COS(RADIANS(Lat1))*SIN(RADIANS(Lat2))-  
SIN(RADIANS(Lat1))*COS(RADIANS(Lat2))*COS(RADIANS(Lon2-Lon1)), SIN(RADIANS(Lon2-  
Lon1))*COS(RADIANS(Lat2))))+360, 360)
```

Function 6

Bearing / Azimuth Calculation

```
=IFERROR(  
  IFS(F2="Current City","No Direction",  
    OR(F2>=337.5,F2<22.5),"N",  
    F2<67.5,"NE",  
    F2<112.5,"E",  
    F2<157.5,"SE",  
    F2<202.5,"S",  
    F2<247.5,"SW",  
    F2<292.5,"W",  
    F2<337.5,"NW"  
  ), "No Direction"  
)
```

Bearing w.r.t. city	
City	Kanpur
Latitude	26.3704694
Longitude	80.2925157

Function 6

Bearing / Azimuth Calculation

Result

State	District	Latitude	Longitude	Distance(kM)	Bearing(degree)	Direction
Andaman and Nicobar	Andaman Islands	12.38257107	92.8229105	2033.296405	137.5152419	SE
Andaman and Nicobar	Nicobar Islands	7.83529075	93.5116012	2489.690775	143.5067795	SE
Andhra Pradesh	Adilabad	19.28451395	78.8132123	802.3473805	191.1863101	S
Andhra Pradesh	Anantapur	14.31206556	77.4601584	1372.8003	192.9402667	S
Andhra Pradesh	Chittoor	13.33109263	78.927639	1456.880598	185.8684596	S
Andhra Pradesh	Cuddapah	14.49028913	78.6961387	1331.394709	187.4702904	S
Andhra Pradesh	East Godavari	16.78271832	82.2432075	1084.95843	168.9130905	S
Andhra Pradesh	Guntur	15.88492577	80.5865761	1166.336497	178.4462147	S
Andhra Pradesh	Hyderabad	17.38971556	78.4666889	1016.177118	191.0366822	S
Andhra Pradesh	Karimnagar	18.51548319	79.4376543	877.8317059	185.9120655	S
Andhra Pradesh	Khammam	17.68392133	80.6700864	966.6816093	177.6193834	S
Andhra Pradesh	Krishna	16.14387275	81.1480506	1140.583276	175.3802049	S
Andhra Pradesh	Kurnool	15.43759715	77.9008684	1240.696083	191.997241	S

Function 7

Conditional Formatting

- Conditional formatting in Excel functions similarly to **attribute-based querying** in GIS. Instead of spatially selecting features, it uses logical conditions to filter and highlight records that meet specific criteria.
- Conditional formatting serves as a practical substitute for GIS attribute queries when working with spatial data in Excel.
- It allows users to filter and highlight specific records such as “Government-owned Clinics in the Ashanti Region” by applying logical AND/OR conditions across multiple attribute columns

Excel Implementation

1. Imported the health facility dataset into Excel, containing fields like Region, District, Facility Name, Type, Ownership, Latitude, and Longitude.
2. Created a separate condition table (columns L–M) with three filter criteria:

Conditions	
Ownership	Government
Type	Clinic
Region	ashanti

Function 7

Conditional Formatting

Excel Formula

`=AND($Region_col=$M$5, $type_col=$M$4,$FOwnership_col=$M$3)`

Where \$M\$5 = Region, \$M\$4 = Type, and \$M\$3 = Ownership reference cells.

1. Selected the data range `A\$2: H\$2757` and opened Conditional Formatting > New Rule.
2. Used a formula-based rule with AND logic to highlight rows matching all three conditions simultaneously:
3. Applied Yellow Fill with Dark Yellow Text to visually distinguish the filtered facilities from the rest of the dataset.
4. Result Interpretation Rows satisfying all three conditions got highlighted.
5. select all the rows(i.e. A:H) then click on Sort and Filter then Custom Sort then in the sort on column drag down to Cell color and then in color/icon column select the light color.

Function 7

Conditional Formatting

Result

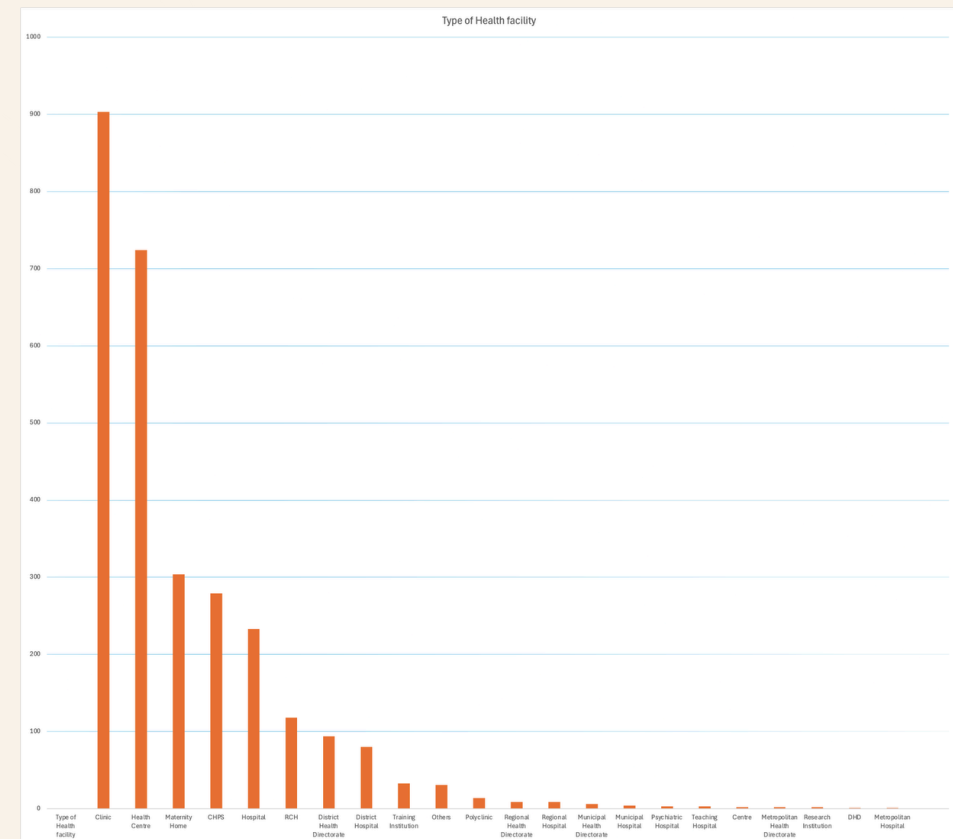
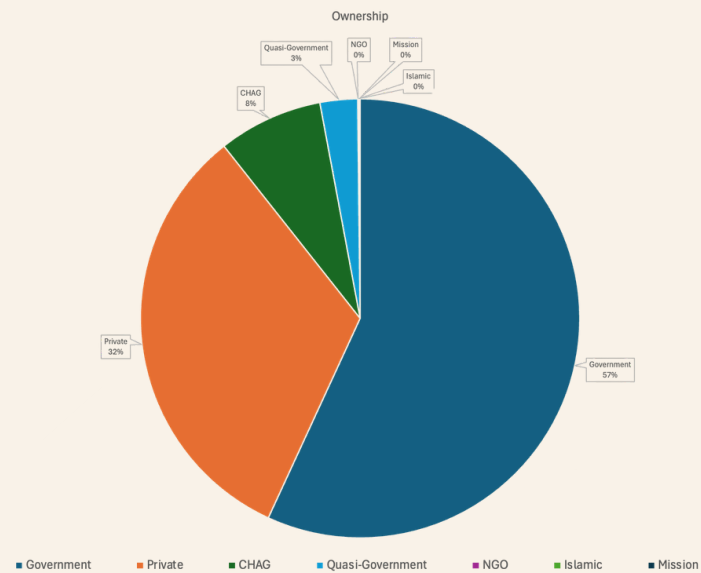
Region	District	FacilityName	Type	Town	Ownership	Latitude	Longitude
Ashanti	Adansi South	Hwidiem Community Clinic	Clinic	Hwidiem	Government	5.89432	-1.46766
Ashanti	Adansi South	Obonsu Clinic	Clinic	Obonsu	Government	5.95324	-1.35225
Ashanti	Adansi South	Atwereboana Clinic	Clinic	Atwereboana	Government	5.96235	-1.34213
Ashanti	Adansi South	Allen Health Clinic	Clinic	Wamasi	Government	6.08691	-1.69553
Ashanti	Obuasi Municipal	Anyinam Clinic	Clinic	Obuasi-Anyinam	Government	6.19897	-1.68205
Ashanti	Amansie Central	Mile 14 Community Clinic	Clinic	Mile 14	Government	6.22126	-1.82139
Ashanti	Amansie Central	Akrofrom Community Clinic	Clinic	Akrofrom	Government	6.39651	-1.60783
Ashanti	Bosome Freho	Gyasiakrom Health Centre	Clinic	Gyasiakrom	Government	6.40145	-1.47313
Ashanti	Bosome Freho	Duasi Community Clinic	Clinic	Duasi	Government	6.46863	-1.39522
Ashanti	Asante-Akim South	Banso SDA Clinic	Clinic	Banso	Government	6.47584	-1.19674
Ashanti	Bosome Freho	Detieso Community Clinic	Clinic	Detieso	Government	6.47674	-1.44168
Ashanti	Kumasi Metropolitan	Sepe-Buokrom Community Clinic	Clinic	Buokrom	Government	6.72118	-1.59593
Ashanti	Ejisu-Juaben Municipality	Tikrom Clinic	Clinic	Tikrom	Government	6.73154	-1.51337
Ashanti	Asante-Akim North Municipal	Nutrition Rehabilitation Centre	Clinic	Konongo	Government	6.79529	-1.08871
Ashanti	Atwima-Nwabiagya	Atwima Clinic	Clinic	Atwima	Government	7	-2
Ashanti	Sekyer Central	Jeduako Clinic	Clinic	Jeduako	Government	7.09501	-1.20917
Ashanti	Ejura Sekyeredumasi	Babaso MCH	Clinic	Babaso	Government	7.34065	-1.32417
Ashanti	Ejura Sekyeredumasi	Ejura MCH	Clinic	Ejura	Government	7.36556	-1.34358
Ashanti	Ejura Sekyeredumasi	Homako Community Clinic	Clinic	Homako	Government	7.42915	-1.45106
Ashanti	Ejura Sekyeredumasi	Zambrama Community Clinic	Clinic	Zambrama	Government	7.52022	-1.42013
Ashanti	Asante-Akim South	Morso Clinic	Clinic	Morso	Government	6.655180000	-1.10681
Ashanti	Kumasi Metropolitan	KMA Clinic	Clinic	Adum	Government	6.684880000	-1.62353
Ashanti	Ejura Sekyeredumasi	Nkwanta Community Clinic	Clinic	Ejura Nkwanta	Government	7.513780000	-1.34725
Brong Ahafo	Asunafo South	Asarekrom Rural Clinic	Clinic	Asarekrom	Government	6.57574	-2.63918
Brong Ahafo	Asutifi	Sienkyemu Rural Clinic	Clinic	Sienkyemu	Government	6.70723	-2.39529
Brong Ahafo	Asunafo North Municipal	Ayomso Rural Clinic	Clinic	Ayomso	Government	6.76364	-2.58248
Brong Ahafo	Asunafo North Municipal	Ayum Forest Clinic	Clinic	Mim	Government	6.90647	-2.56449
Brong Ahafo	Asutifi	Gambia No1 Rural Clinic	Clinic	Gambia No.1	Government	7.0378	-2.6357
Brong Ahafo	Dormaa Municipal	Kwameasua Presby Clinic	Clinic	Kwameasua	Government	7.16715	-2.7701
Brong Ahafo	Dormaa Municipal	Asikasu Rural Clinic	Clinic	Asikasu	Government	7.1893	-2.89066
Brong Ahafo	Dormaa Municipal	Amasu Rural Clinic	Clinic	Ammasua	Government	7.22802	-2.95807
Brong Ahafo	Tano North	Adrobaa Rural Clinic	Clinic	Adrobaa	Government	7.2628	-2.13179
Brong Ahafo	Dormaa East	Kyeremasu Rural Clinic	Clinic	Kyeremasu	Government	7.26312	-2.76114

Function 8

Pictorial Representation

Formulas Used

=COUNTIF(D2:D2857,"CLINIC")



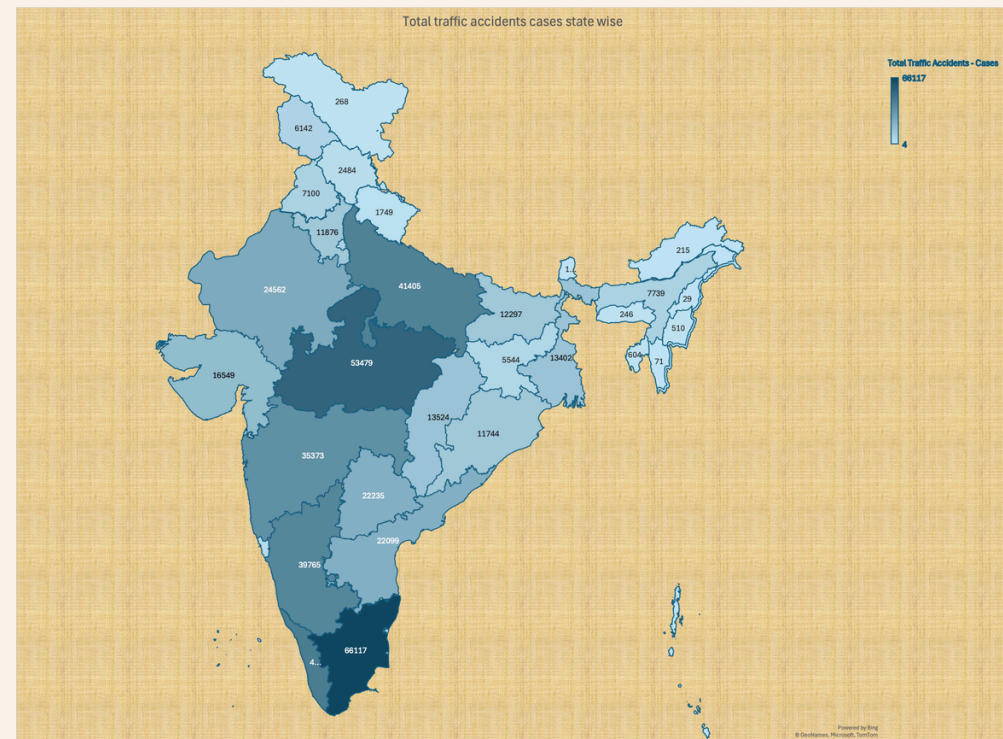
Function 9

Choropleth Map using Excel 3D Maps (Bing Maps)

A choropleth map is a thematic map where regions are shaded in proportion to the value of a statistical variable.

In the below map , total traffic accident cases per state are shown.

Darker shades indicate higher values and lighter shades indicate lower values.



Function 10

Coordinate Mapping using ArcGIS for Excel(Add-in)

Coordinate mapping is the process of plotting geographic point locations on a map using their latitude and longitude values. ArcGIS for Excel is an Esri add-in that enables spatial visualization of tabular coordinate data directly within the Excel environment using interactive Esri basemaps.

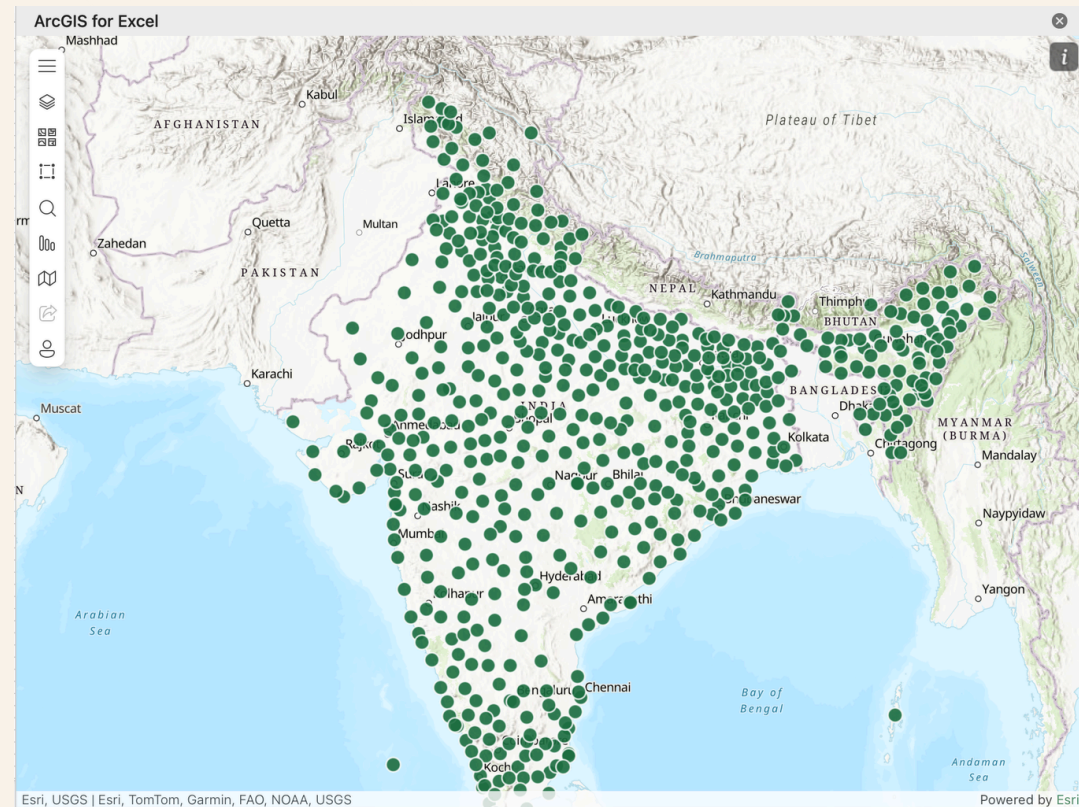
Excel Implementation

1. Tabular data containing State, District, Latitude and Longitude columns was arranged in Excel. Each row represented one district with its corresponding centroid coordinates.
2. ArcGIS for Excel was enabled through Insert > Add-ins > Get Add-ins from the Excel ribbon and signed in as a Guest.
3. The coordinate columns were selected and chosen into the ArcGIS map panel, specifying which columns represent latitude and longitude fields.
4. The add-in rendered all district points as circular markers over an Esri world basemap, with India's boundary clearly visible and all state clusters identifiable.
5. Marker size, color, and basemap style were adjusted for better visual clarity and presentation and map extent was locked using navigation tools in toolbars.

Function 10

Coordinate Mapping using ArcGIS for Excel(Add-in)

Result



Function 11

Heat Map Visualization

A visualization technique where data values are represented using color variations instead of numbers

										Railway Crossing									
Sl. No.	State	Road Accidents	Road Accidents	Road Accidents	Roadway Accidents	Railway Accidents	Roadway Accidents	Roadway Accidents	Roadway Accidents	Railway Crossing	Total Traffic Accidents - Cases	Total Traffic Accidents - Injured	Total Traffic Accidents - Died						
1	Andhra Pradesh	21070	21340	8293	1029	0	1037	0	0	0	22099	21340	9330						
2	Arunachal Pradesh	215	177	152	0	0	0	0	0	0	215	177	152						
3	Assam	7028	5679	3060	689	0	693	22	0	22	7739	5679	3775						
4	Bihar	10801	7068	8898	1444	6	1446	52	0	53	12297	7074	10397						
5	Chhattisgarh	13091	11459	5890	429	21	429	4	0	4	13524	11480	6323						
6	Goa	3012	1071	274	54	0	54	0	0	0	3066	1071	328						
7	Gujarat	15777	15139	7634	766	0	777	6	0	6	16549	15139	8417						
8	Haryana	10654	8353	5228	1127	26	1101	95	0	95	11876	8379	6424						
9	Himachal Pradesh	2484	3891	979	0	0	0	0	0	0	2484	3891	979						
10	Jharkhand	5175	3747	3898	362	19	346	7	0	7	5544	3766	4251						
11	Karnataka	39765	48154	11705	0	0	0	0	0	0	39765	48154	11705						
12	Kerala	43592	49317	4317	194	0	194	184	1	185	43970	49318	4696						
13	Madhya Pradesh	51965	51264	13918	1275	0	1275	239	0	239	53479	51264	15432						
14	Maharashtra	30546	23878	15748	4827	1686	3145	0	0	0	35373	25564	18893						
15	Manipur	508	817	127	2	0	2	0	0	0	510	817	129						
16	Meghalaya	246	310	162	0	0	0	0	0	0	246	310	162						
17	Mizoram	71	28	83	0	0	0	0	0	0	71	28	83						
18	Nagaland	24	42	22	0	0	0	5	0	5	29	42	27						
19	Odisha	11663	10269	5470	27	0	27	54	54	0	11744	10323	5497						
20	Punjab	6122	3372	4688	978	74	914	0	0	0	7100	3446	5602						
21	Rajasthan	23620	22249	11103	942	8	943	0	0	0	24562	22257	12046						
22	Sikkim	199	371	98	0	0	0	0	0	0	199	371	98						
23	Tamil Nadu	64105	67703	17884	2012	189	1833	0	0	0	66117	67892	19717						
24	Telangana	21619	20209	7559	616	0	625	0	0	0	22235	20209	8184						
25	Tripura	575	541	241	27	0	27	2	0	2	604	541	270						
26	Uttar Pradesh	36875	21696	24109	3190	35	3166	1340	4	1340	41405	21735	28615						
27	Uttarakhand	1674	1613	1042	74	0	74	1	0	1	1749	1613	1117						
28	West Bengal	10635	8124	5788	2221	286	1935	546	53	493	13402	8463	8216						
Total (States)		433111	407881	168370	22285	2350	20043	2557	112	2452	457953	410343	190865						

Discussion

Applications

- Distance Calculation can be used in various types of analysis such as pipeline route optimization, highway planning, rail or metro corridor analysis etc.
- Nearest neighbor is used to planning of facility which would be required by us in the near future such as (nearest hospital, fire station, police station, railway station, school etc.).
- For road and highway design, drainage channel design (minimum slope for flow), and erosion susceptibility mapping.
- Survey traverse calculations, road alignment bearing determination, pipeline routing, and military/civil defense mapping.
- Pictorial representation helps in planning infrastructure by identifying dominant facility types in a region, which one must keep in consideration regarding the facility.
- Identifying accident-prone highway corridors for road safety intervention.
- By using ArcGIS Add-in, we can do Quick spatial verification of survey coordinate datasets. We can do Preliminary site mapping before detailed GIS analysis.

Discussion

Limitations

- For distance calculation here we are assuming a perfect sphere. For high-accuracy engineering surveys, the WGS84 ellipsoid model (used in QGIS/ArcGIS) gives more precise results, especially over long distances.
- In excel we are computing slope by just two discrete points where as a true slope map requires raster DEM processing across thousands of grid cells simultaneously, which surely requires ArcGIS or QGIS Raster Analysis.
- We need to have an active internet connection to use ArcGIS. There is no support provided for shapefile or raster data integration.
- Pictorial representations are non-spatial and they show counts but cannot display geographic distribution.
- We can find the nearest neighbor among a finite set of points in excel, but cannot perform true k-d tree spatial indexing or handle thousands of candidates efficiently.

Conclusion

- Why dedicated GIS tools exist in the first place?
- The Haversine formula, bearing calculations, slope estimation, and proximity queries all translated surprisingly well into Excel's formula environment.
- What we came away with is a clearer sense of where Excel genuinely fits in a GIS workflow
 - early-stage data verification
 - rapid proximity checks
 - attribute filtering

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Thank

You



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